

# Cayman Roden

## AI Engineer - Production LLM Systems - RAG - Evals

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Proof links: [DocExtract demo](#) | [EnterpriseHub demo](#) | [PyPI: mcp-server-toolkit](#)

Open source: [LiteLLM PR #24551](#) | [FastAPI PR #15217](#) | [AI red-team portfolio](#)

## Professional Summary

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AI engineer with 3+ years building production LLM systems on contract. I ship RAG pipelines, multi-agent orchestration, and LLM evaluation gates that block regressions in CI before deploy.

Proof: DocExtract (95.5% extraction F1 on a CI-replayed 28-case golden baseline, pgvector RAG, shipped prompt-injection defense), a 7-agent CRM orchestration platform, a PyPI-published MCP framework, and a 3-month client engagement that processed 500+ real leads. Stack: Python, FastAPI, pgvector, model routing, OpenTelemetry, LLM-as-judge evaluation.

## Experience

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### AI Engineer | Independent Contract | Remote | June 2022 - Present

Contract AI engineering for small-business clients. Built and shipped the production systems detailed below (DocExtract, EnterpriseHub, mcp-server-toolkit) plus a real-time voice pipeline, a multi-tenant chat widget, and a workflow automation API, each with eval-gated CI and production handoff documentation.

#### Acuity Real Estate (client engagement) | Jan 2026 - Mar 2026

- Built and shipped a Claude-powered bilingual (EN/ES) SMS lead-qualification system: 500+ inbound leads processed with zero reported incidents over the 3-month production run, with CRM auto-updates for warm-lead routing.
- Worked directly with the broker to define qualification rubrics and escalation rules; built 3 specialized SMS bots (Lead Intake, Buyer Qualification, Seller Qualification) with cross-bot routing, admin handoff, and GoHighLevel CRM integration (contact tagging, pipeline automation).
- Designed an LLM-as-judge golden evaluation set with adversarial fixtures (contradictory documents, truncated inputs, multi-language, high-noise OCR) scored against a rubric with evidence extraction.
- Delivered 1,700+ tests at handoff with a formal 11-section test-strategy document, coverage report, runbook for new flows, and an audit of 226 existing GHL workflows. Zero production regressions.

### Service Industry and Operations | Los Angeles and Palm Springs, CA | 2010 - 2024

Customer-facing service, sales, and bookkeeping roles held while building the AI engineering practice above. Named employers, dates, and references available on request.

## Flagship Production Systems

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**DocExtract** - [repo](#) | [live demo](#) Document-extraction RAG service. 95.5% accepted extraction F1 on a 28-case golden baseline, replayed deterministically by an offline CI job at zero API cost; full 72-case corpus (51 golden + 21 adversarial) committed. Two-pass Claude extraction with tool-use verify, pgvector retrieval, Gemini Flash LLM-as-judge CI gate. Haiku classification routing is ~12x cheaper per token than Sonnet, with a two-sample z-test A/B framework (model\_ab\_test.py) wired to validate quality parity. Cost ~\$0.03/doc, p95 ~4.1s, ~88% straight-through (modeled from a committed pricing and metering rig). 1,280 tests, 81% coverage.

**EnterpriseHub** - [repo](#) | [live demo](#) 7-agent multi-CRM orchestration mesh (~10 with auto-discovery) with 4-dimension weighted routing (success, load, cost, latency), circuit-breaker failover, an hourly spend kill-switch, and an eval-gated CI on a 50-case golden set. FastAPI, PostgreSQL, Redis, LangGraph.

**mcp-server-toolkit** - [PyPI](#) | [repo](#) Published MCP server framework: JWT/OAuth 2.1 (HS256/RS256/JWKS), OpenTelemetry tracing with per-span cost, a multi-tier cache layer (sub-microsecond L1 hits, committed micro-benchmark in benchmarks/RESULTS.md), 9 pre-built servers, and a 30-case adversarial injection corpus. 600 tests, ~83% coverage, Python 3.10-3.12 CI matrix.

## AI Security and Evaluation

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- Designed and shipped an indirect prompt-injection defense for untrusted document text (ADR-0020): instruction-hierarchy separation, untrusted-data fencing, an injection scanner, and output sanitization that

strips exfiltration keys so a successful injection still cannot leak secrets, the system prompt, or downstream records.

- Built a 72-case golden-plus-adversarial eval corpus (prompt injection, PII leak, hallucination bait) with an offline CI replay pipeline that gates merges deterministically at zero API cost.
- Maintain pre-registered injection studies with falsifiers and Wilson 95% CIs, a Lakera Gandalf CTF walkthrough, and public hypothesis retractions when the data did not support the claim (portfolio link in header).

## Additional Engineering

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- Real-time voice pipeline (FastAPI + WebSocket): Silero VAD, Deepgram Nova-3 STT, Claude Sonnet streaming, Deepgram Aura-2 TTS; barge-in interrupt, per-component latency tracking (P50/P95/P99 via /api/metrics/latency), 171 pytest cases gating CI.
- Multi-tenant embeddable chat widget (FastAPI + pgvector + Claude + Next.js, Stripe billing, per-tenant isolation) and a YAML-driven workflow automation API (Redis async workers, SSE progress, 148 tests). GitHub Actions CI across 16 AI repos with per-repo coverage gates, ruff, mypy, and security scans; eval-regression gates in 3 (DocExtract, EnterpriseHub, mcp-server-toolkit).

## Open Source

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- [LiteLLM PR #24551](#) (BerriAI/litellm, 47K+ stars, open): typed exception mapping for Router fallback so AuthenticationError, RateLimitError, and NotFoundError surface distinctly; tests for 4 error cases.
- [FastAPI PR #15217](#) (fastapi/fastapi, 98K+ stars, open): documentation warning about the BackgroundTasks and Response.background interaction.

## Technical Skills

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**AI/LLM:** Claude API | OpenAI | Gemini | Retrieval-Augmented Generation (RAG) | Semantic Search | Embeddings | Vector Search | Vector Database | pgvector | LangChain | LangGraph | MCP | Multi-Agent | Tool Calling | LiteLLM | Structured Output | JSON Schema | Prompt Engineering | Model Evaluation | LLM-as-judge | Faithfulness | RAGAS

**Backend:** Python | FastAPI | SQL | PostgreSQL | Redis | SQLAlchemy | Pydantic | Docker | Terraform | ARQ | Render | Vercel | Cloud Run | AWS | GCP | REST APIs | GitHub Actions CI/CD | pytest | Test-Driven Development (TDD) | TypeScript | Next.js | React

**Systems:** Async | SSE | WebSockets | Prompt Caching | Cost Attribution | Model Routing | OpenTelemetry | Prometheus | Grafana | LLM Evals | Golden Sets | Regression Suites

## Education and Continuous Learning

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11 certificates, ~750 hours (Google, Microsoft, IBM, Duke, Vanderbilt, Linux Foundation, DeepLearning.AI):

- Duke LLMOps Specialization
- Vanderbilt Prompt Engineering
- IBM Generative AI Fundamentals
- Google Cloud Generative AI Leader
- Linux Foundation OSS / Linux / Git

Self-trained engineer; all coursework verifiable on request.